

UNIVERSITY OF ZAGREB
FACULTY OF ELECTRICAL ENGINEERING AND COMPUTING

MASTER THESIS No.

**DEEP LEARNING-BASED ANIMAL
RE-IDENTIFICATION FOR NON-INVASIVE
WILDLIFE MONITORING AND
CONSERVATION**

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Zagreb, June, 2025

SVEUČILIŠTE U ZAGREBU
FAKULTET ELEKTROTEHNIKE I RAČUNARSTVA

DIPLOMSKI RAD br.

**DUBOKO UČENJE ZA PONOVRNU
IDENTIFIKACIJU ŽIVOTINJA U
NEINVAZIVNOM PRAĆENJU DIVLJIH VRSTA I
OČUVANJU PRIRODE**

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The master thesis assignment text in English goes here.

Ovdje dolazi tekst zadatka diplomskog rada na hrvatskom jeziku.

Thanks for the tea...

Contents

1	Introduction	4
2	Related Work	6
2.1	Traditional Computer Vision Approaches	6
2.2	Species-Specific Trait-Based Re-Identification	7
2.3	Deep Learning-Based Methods	7
2.4	Hybrid Approaches with Learned Local Features	8
2.5	Key Challenges and Limitations	8
3	Datasets	9
3.1	WildlifeReID-10k	9
3.1.1	Train/Test Leakage Issues and Split Protocols	10
3.1.2	MegaDescriptor Training Leakage	10
3.1.3	Datasets Overview	11
3.1.4	Class balance visualizations	11
4	Methodology	13
4.1	Preprocessing	13
4.1.1	Mantiuk Tone Mapping	13
4.1.2	Segmentation with Grounded SAM2	13
4.2	Feature Extraction	14
4.2.1	Why combine global and local features?	14
4.2.2	Global Embeddings	14
4.2.3	Local Features	15
4.2.4	Local Matching	16
4.3	Feature Aggregation and Representation	16

4.3.1	Training PCA and GMM models	17
4.3.2	Fisher Vector Computation	18
4.3.3	Parameter Choices and Rationale	19
4.4	Geometric Verification	19
4.4.1	Theoretical Foundation	19
4.4.2	Establishing Initial Correspondences	20
4.4.3	Transformation Estimation	20
4.4.4	Scoring and Verification	20
5	Classification	22
5.1	Problem Formulation	22
5.2	Overview: a 3-tier funnel	22
5.3	Tier 1: Candidate Retrieval	23
5.4	Tier 2: Score Calibration and Fusion	24
5.4.1	Calibration Set Construction	24
5.4.2	Calibrators and Fusion Rules	24
5.5	Tier 3: Geometric Verification Reranking	25
5.6	End-to-end Inference	25
6	Population Size Estimation	27
6.1	Introduction	27
6.2	Theoretical Foundation	28
6.2.1	Population Size as Connected Components	28
6.2.2	Two-Stage Estimation: Nested Monte Carlo and Nested Importance Sampling	28
6.3	HITL-NIS Implementation	29
6.3.1	Graph and Annotator Abstraction	29
6.3.2	Cheap Proposal Signals: Global Embeddings and Fisher Vectors	30
6.3.3	Calibrating Scores to Probabilities for Sampling	30
6.3.4	Outer Sampling Choice	31
6.3.5	Complexity and Diagnostics	31
6.4	Human Labeling Error	31
6.4.1	Explicit Noise Modeling	32

6.4.2	Confirmation by Repeated Voting	32
6.5	Parameter Configuration	32
7	Experiments and Results	34
8	Conclusion and Future Work	35
	References	36
	Abstract	40
	Sažetak	41

1 Introduction

The conservation and effective management of wildlife populations depend crucially on accurate methods for monitoring animal abundance and diversity. Traditional approaches, including manual identification, tagging are often invasive, costly, time-consuming and impractical at large scales or in remote environments. In recent years, significant progress has been achieved through the application of automated computer vision methods for wildlife monitoring, particularly though individual animal re-identification (Re-ID). Animal Re-ID, which refers to the task of identifying and matching individual animals across multiple observations solely on visual appearance, has emerged as a viable non-invasive method for ecological studies and conservation efforts.[1] Despite the advancements, current visual Re-ID systems still face substantial challenges, especially when applied to real-world scenarios characterized by significant appearance variations, environmental clutter, occlusions, and limited training data. Current methods typically require extensive manual labeling effort or species-specific model training, restricting their applicability in general ecological monitoring and conservation task so developing an efficient, flexible Re-ID framework capable of generalizing to virtually any animal species without requiring significant manual annotation or retraining represents an important and currently unmet need in conservation technology. The main aim of this thesis is just that, to develop a generalizable, robust animal Re-ID and population estimation system capable of performing reliably across diverse animal species with minimal manual intervention. Specifically, this thesis attempts to create a method suitable for real-world scenarios characterized by limited labeled data, eliminating the requirement for extensive manual labeling which can often be very difficult and error-prone cause it requires a certain expertise. To achieve these goals, the research was structured around two interlinked objectives:

1. **Evaluation and optimization of robust classification methodologies:** Before addressing the complex task of population estimation, it was essential to establish a strong foundation for animal Re-ID that generalizes effectively across species. Publicly available datasets, notably WildlifeReID[2], were leveraged for evaluation and testing. This phase involved extensive experimentation with robust feature extraction methods, including DISK[3] and LightGlue[4] combined with dimensionality reduction and aggregation approaches such as PCA, GMM and Fisher Vectors.
2. **Development of population size estimation method:** Following the establishment of a reliable and generalizable individual classification framework, the ultimate goal was population counting. Implementation of the Nested Importance Sampling (NIS)[5], a human-in-the-loop statistical framework leveraging pairwise similarity scores produced by the general-purpose Re-ID model. Critically, this method requires minimal human input and does not require additional training or extensive labeling for new animal species or datasets, thus significantly reducing manual labor and expertise requirements in practical scenarios. Several potential weak points were discovered in the paper itself which were attempted to address in this thesis.

To further validate the practicality and effectiveness of the developed methods in real-world conditions, original data was collected specifically for this thesis. Two trail cameras were deployed on Medvednica mountain for approximately five months, gathering data of local wildlife populations (mostly wild boars and deer). The final and crucial stage of this thesis involved applying the developed Re-ID and population estimation framework directly to this newly collected, unlabeled dataset.

2 Related Work

In this chapter, we'll review existing approaches to visual animal re-identification (Re-ID) with an emphasis on methods intended to generalize across species. The literature will be organized into thematic categories, from traditional computer vision techniques to modern deep learning methods, including hybrid methods that combine the two. We also discuss key challenges in the field, such as limited data, pose variation and open-set identification. We'll synthesize the findings from prior work and emphasize the gaps that this thesis is trying to address.

2.1 Traditional Computer Vision Approaches

Early animal Re-ID systems relied on hand-crafted visual feature and local descriptors. A prime example is HotSpotter[6], a SIFT-based[7] algorithm that identifies individuals by comparing local keypoints in images. HotSpotter uses viewpoint-invariant SIFT descriptors and scores matches by emphasizing the most distinctive keypoints (the titular "hot spots") on an animal's coat pattern. [8] compared SIFT, SURF and SuperPoint on identifying sunfish individuals and demonstrated that the number of matching keypoints between a pair of images is a strong indicator for the likelihood that they contain the same individual. These traditional local-feature methods do not require training on identity labels and are, by definition, species-agnostic, making them applicable to a wide variety of species. HotSpotter and similar algorithms have been successfully used on numerous species with distinctive markings, including zebras, giraffes, jaguars, ocelots and leopards.[9] The drawback, however, is that purely hand-crafted methods cannot fully exploit any available training data since their matching heuristic is fixed and may miss more subtle or global cues.

2.2 Species-Specific Trait-Based Re-Identification

Another line of research focuses on species-specific visual traits for re-identification. Instead of analyzing the entire image or generic local patches, these approaches isolate particular biometric features unique to the species at hand, for example the ear shapes of elephants, the outline of a shark’s dorsal fin, or the contour of a whale’s tail (fluke), and use them for re-identification. Both traditional image analysis and deep learning have been employed to recognize such traits. For example, algorithms such as CurvRank[10] and finFindR[11] compute and match the shapes of whale flukes or shark fins to identify individuals. These methods often don’t require individual identity labels for training, rather a detector or feature extractor for the trait can be trained using only that trait’s annotations, then match those features between images. The advantage of trait-specific methods is that they leverage highly distinctive features (when they are present) and can work with minimal ID labeling. However, a major limitation is that they are species-specific so each system is tailored for identifying only one species and usually it can’t be transferred to other species. This lack of generality is a significant barrier to wider use as it requires developing a new method for each studied species. This thesis is motivated by overcoming this limitation by developing a method, based on existing research, that does not depend on any single species’ unique trait.

2.3 Deep Learning-Based Methods

With rise of deep learning, a number of methods have been proposed that learn features directly from image data, without manual feature engineering. Modern deep learning approaches for animal re-ID typically employ CNNs (Convolutional Neural Networks) or related architectures to produce an embedding (feature vector) for each image, such that images of the same individual are close in feature space while those of different individuals are far apart. These can be trained either as classification (training a CNN to classify a fixed set of known individuals) or via metric learning (training the network to differentiate between same-individual vs. different individual image pairs). In general, metric learning approaches are favored for wildlife re-ID because they naturally handle open-set problems. A classification-based model treats re-ID as identifying one of N known individuals, which fails when a new individual outside the training set appears.

By contrast, metric-based methods learn a similarity function and can thus match new individuals by embedding comparison, without needing to retrain for new identities.[9]

2.4 Hybrid Approaches with Learned Local Features

A promising direction in animal re-id is to combine the strengths of traditional and deep learning methods into a hybrid pipeline. The idea is to use modern learned features from deep networks in place of the hand-crafted ones, but still employ them in a local matching and aggregation framework as in traditional approaches. A good example of this is the ALFRE-ID[9] pipeline (Aggregated Local Features for Re-Identification). ALFRE-ID uses a CNN extractor to extract local patch descriptors from an animal's coat pattern, then aggregates these local features into a global descriptor for the image.

2.5 Key Challenges and Limitations

Research in animal re-identification encounters several fundamental challenges that limit real-world performance. Key issues identified in the literature are:

Limited labeled data. Collecting individual-level labels is labor-intensive, so datasets often contain only a handful of images per subject, leading to overfitting and poor generalization.

Pose variation. Animals appear under diverse viewpoints and behaviors, drastically altering their visual features. Identification accuracy drops when profiles or unusual poses replace frontal views.

Occlusions. Vegetation, group interactions, or partial camera-trap captures often conceal distinguishing markings, hindering feature extraction.

Open-set identification. Field deployments must handle individuals unseen during training; closed-set models misclassify such cases. Open-set recognition methods highlight the need to reject unknown identities.

3 Datasets

Accurate and generalizable animal re-identification (Re-ID) system depends on the quality and diversity of datasets used for development and evaluation. This thesis evaluates the proposed approach on publicly available datasets, mostly from WildlifeReID-10k dataset collection[12]

3.1 WildlifeReID-10k

WildlifeReID-10k[12] is a benchmark dataset collection explicitly curated for evaluating animal Re-ID methods across a variety of species. The primary goal of this dataset is to enable systematic benchmarking of general-purpose animal Re-ID methods. It aggregates 37 existing datasets into a unified format to enable easy evaluation. In total, WildlifeReID contains 140,488 images of 10,772 unique individuals spanning 33 species. The methods described in this thesis were tested on 7 datasets from that collection: ATRW (tigers), CowDataset (cows), ElPephants (elephants), CTai (chimps), Chicks4FreeID (chickens), SealID (seals) and SeaStarReID2023 (sea stars). They were chosen because they provide the range of challenges of re-identification and the testing on all 37 datasets would be too computationally expensive.

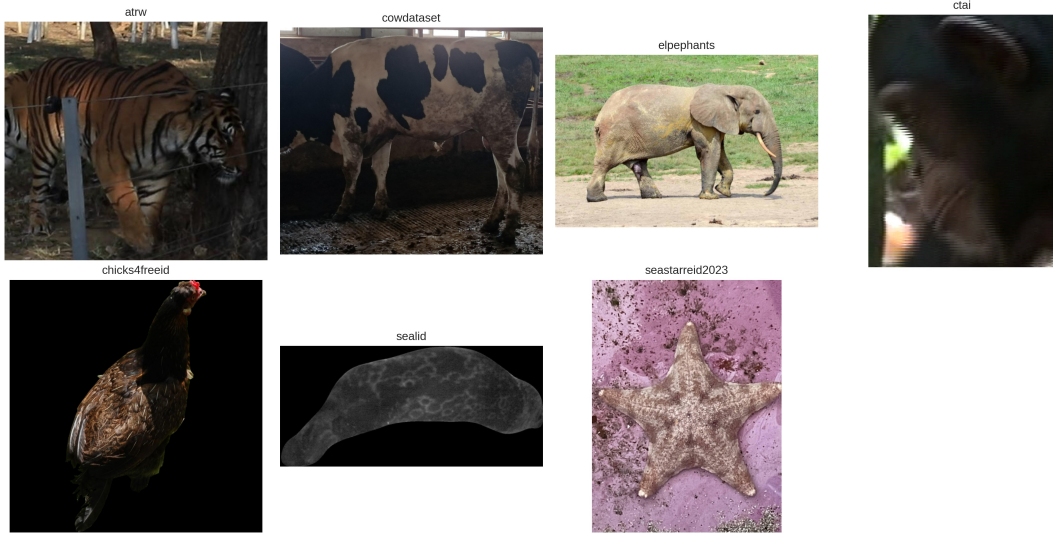


Figure 3.1: Example images from the datasets used in this thesis.

3.1.1 Train/Test Leakage Issues and Split Protocols

A central (and often not addressed) issue in animal Re-ID evaluation is train-to-test leakage caused by visually almost identical images (for example, consecutive video frames or images from camera traps taken in very quick succession during a single observation) ending up in both train and test sets. There are also mistakes in the datasets, so in some cases there are literal duplicates present. Because of this random splits often result in overly optimistic results that do not reflect real-world generalization.[12]

To address this, WildlifeReID-10k proposes two split strategies[12]:

- **Time-aware splits:** some of the datasets come with timestamps and this is used to split the images chronologically per identity using a cut-off date so that images from the same encounter do not leak across splits
- **Similarity-aware splits:** when timestamps are not available, an embedding model DINOv2[13] is used to cluster the images, leveraging the fact that near duplicates should lie close in the embedding space

3.1.2 MegaDescriptor Training Leakage

In this thesis I used the MegaDescriptor-L-384 (MD) model for extracting of global embeddings from images, this model was trained on a subset of datasets from WildlifeReID-

10k so for those datasets in evaluation I disregarded the time-aware/similarity-aware splits and used the exact same splits that the model was trained on to avoid additional data leaks. Those datasets are: ATRW, CTai and SealID and in later parts of the thesis will be marked with * to signify this. The Exception to this is the ELPephants[14] dataset which uses a random split.

3.1.3 Datasets Overview

Dataset	# images	# individuals	ratio	split type
ATRW*	5,415	182	29.8	Original MD split
CTai*	4,662	71	65.7	Original MD split
SealID*	2,080	57	36.5	Original MD split
CowDataset	1,485	13	114.2	Time-aware split
Chicks4FreeID	1,146	50	22.9	Similarity-aware split
SeaStarReID2023	2,187	95	23.0	Time-aware split
ELPephants	2078	276	7.5	Random split

Table 3.1: Datasets used for evaluation in this thesis.

3.1.4 Class balance visualizations

Another issue with Re-ID datasets is that they are typically highly imbalanced.



Figure 3.2: Class balance (number of images per individual) for the datasets used.

4 Methodology

This chapter describes the core stages of the proposed animal re-identification pipeline, used in both the classification mode and counting mode: preprocessing, feature extraction (global and local), aggregation of local descriptors into Fisher vectors (FV) and geometric verification used as an additional consistency check and for scoring.

4.1 Preprocessing

Wildlife images are rarely captured under controlled conditions, illumination can vary significantly, there can be background clutter and/or occlusion. Preprocessing is used to reduce the variation and potential background bias.

4.1.1 Mantiuk Tone Mapping

Tone mapping compresses the dynamic range and enhances local contrast, making subtle patterns on the animal body more visible. Following the recommendation in [9], this thesis uses Mantiuk tone mapping as an optional preprocessing step.

4.1.2 Segmentation with Grounded SAM2

Even if the images in a dataset are cropped, background can still dominate local or global feature extraction which can lead to background bias. To eliminate this effect, a segmentation-based pipeline is used to isolate the animal from the background. The pipeline first localizes the animal using a text prompted detector (GroundingDINO)[15] and then produces a pixel level mask using Segment Anything Model 2 (SAM2)[16]. The mask is applied with soft edges to avoid introducing sharp boundaries that could create non-existent keypoints.



Figure 4.1: Overview of the segmentation-based preprocessing pipeline used for background removal.

Segmentation was not applied to all datasets in this thesis since some already came segmented or, in case of CTai, were already cropped so tightly that it was deemed unnecessary.

4.2 Feature Extraction

Feature extraction is a critical component of animal Re-Identification pipelines, converting raw image data into a representation that can be compared across observations. This thesis uses two complementary families of features: global embeddings and local features.

4.2.1 Why combine global and local features?

Global embeddings summarize the entire image into a single vector and provide a fast, strong retrieval signal. Local features, on the other hand, represent an image as a set of keypoints and descriptors, which can capture fine-grained details and allow geometric consistency checks. These two feature families are evaluated both independently and in combination in later experiments.

4.2.2 Global Embeddings

Global embeddings are a fixed-length feature vector produced by a pre-trained neural network (MD). Two images can then be compared by measuring the distance or similarity between their embeddings (in this thesis, cosine similarity is used).

For global embeddings, I used MegaDescriptor-L-384[2], a large Swin Transformer model trained specifically for animal re-identification. It was trained on 29 publicly

available datasets from WildlifeDatasets collection[2]. This provides a strong baseline similarity signal that is efficient and fast to compute and can be fused with the results of local methods.[17]

4.2.3 Local Features

Local features model an image as a set of keypoints and associated descriptors. Matching local features between two images can give a similarity signal and also enables geometric verification. In practice, this is a two-step process: first keypoints are detected, and then each keypoint is described with a feature vector.

In this thesis I used modern learning-based local feature extractors: SuperPoint[18], DISK[3] and ALIKED[19]. While each of those works well on it's own, experiments have proven that they complement eachother so they have been used as an ensamble. These methods were chosen because they work well out-of-the-box, requiring no additional training or fine tuning and also learning-based detectors generally perform better than traditional ones like SIFT.[7]

Once keypoints are detected, we compute descriptors that summarize the appearance of the local region around each keypoint. An effective descriptor should be discriminative (distinguishing one region from another) and robust (invariant to geometric transformations).

SuperPoint

SuperPoint[18] uses a single convolutional network (CNN) with two heads: one head produces a per-pixel interest-point score map and the other produces a per-pixel descriptor map. It is trained without manual keypoint labels by warping images with random homographies and enforcing that detections and descriptors remain consistent under those warps.

DISK

DISK (DIScrete Keypoints)[3] is an end-to-end trainable local feature framework that integrates detection and description. It is trained to maximize correct matches across image pairs. It outputs keypoint heatmap and a 128-D descriptor vector.

ALIKED

ALIKED is a learning-based local feature extractor that provides keypoints

4.2.4 Local Matching

Local descriptors are compared by matching keypoints between images. In this thesis, matching is performed with LightGlue[4]. Concretely, for each image pair LightGlue is provided with the extracted keypoint coordinates and their descriptors (from Super-Point/DISK/ALIKED) and it outputs a set of tentative matches (index pairs of matching keypoints). We then use those matches as input to geometric verification (Section 4.4).

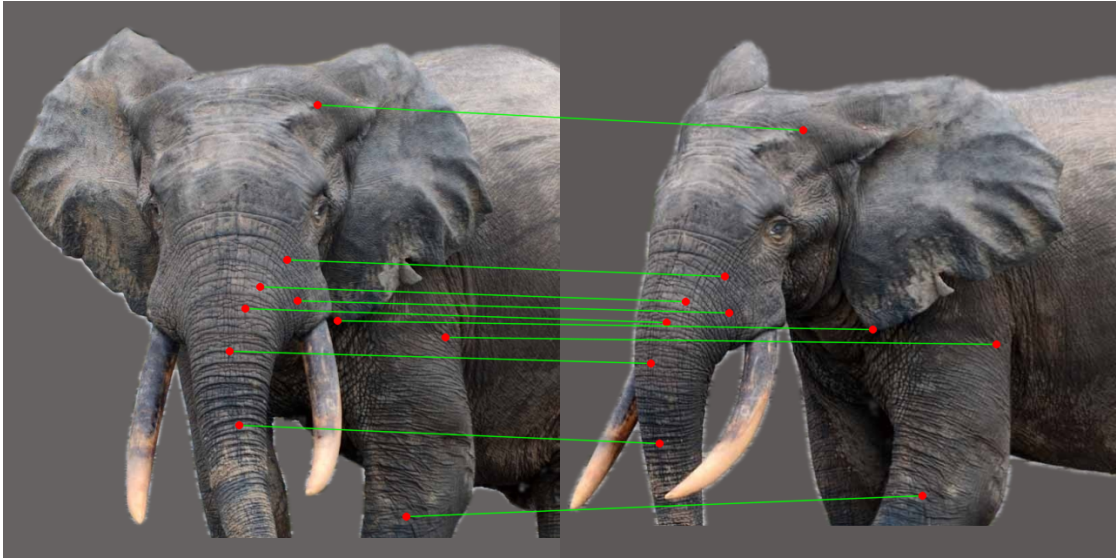


Figure 4.2: Example local feature matches between two images with backgrounds removed. Limited to 10 keypoints for visual clarity.

4.3 Feature Aggregation and Representation

Local feature matching is powerful but computationally expensive when comparing many query images against a large database. To enable efficient large-scale comparison, this thesis also aggregates sets of local descriptors into a single fixed-length representation using Fisher Vectors (FV)[20]. The FV representation is built on PCA (Principal Component Analysis) for dimensionality reduction and GMM (Gaussian Mixture Model) that serves as a visual vocabulary[9].

4.3.1 Training PCA and GMM models

Feature descriptors extracted from images can be high-dimensional and noisy. Thus, dimensionality reduction and normalization are performed to improve efficiency and accuracy.

Initially PCA is applied to decorrelate and reduce the dimensionality of the descriptors. PCA projects the original descriptors onto a lower-dimensional subspace defined by the principal eigenvectors of the covariance matrix computed from the descriptors. This is also important later on for the computation of fisher vectors because they produce very large descriptors.[9]

After dimensionality reduction, a Gaussian Mixture Model (GMM) is trained on the reduced descriptors. A GMM models the descriptor distribution as a weighted sum of K Gaussian components, each parameterized by mean vectors $\{\mu_k\}$, covariance matrices $\{\Sigma_k\}$, and weights $\{\pi_k\}$. Specifically, a GMM is defined as:

$$p(x) = \sum_{k=1}^K \pi_k \mathcal{N}(x|\mu_k, \Sigma_k), \quad (4.1)$$

where $\mathcal{N}(x|\mu_k, \Sigma_k)$ denotes the probability density function of the k -th Gaussian component. This codebook (the trained GMM) captures the visual vocabulary and statistical distribution of the reduced descriptors from the training images.[9]

In simpler terms, PCA finds the directions in which the descriptor values vary the most and keeps only those, so each feature becomes a shorter, uncorrelated vector that is faster to store and compare. Then we model how these compact vectors are distributed. GMM model assumes that the whole set of vectors can be grouped into a number of smooth clusters, each cluster described by its mean, spread and relative weight in the mixture. Together, these clusters form a codebook meaning that every new descriptor can be represented as a weighted combination of the nearest clusters, giving us both a compressed encoding and a way to compute similarity for matching or classification which is of crucial importance in re-identification task.

4.3.2 Fisher Vector Computation

Fisher vectors (FV) are an image representation technique that captures statistical deviations of local features from a visual vocabulary (codebook) modeled by GMM. FV measures how features differ from typical patterns in a reference GMM and they encode both first-order differences (mean shifts) and second-order differences (variance changes) in feature distribution. The output is a high-dimensional vector that acts as a "signature" of an image's unique feature distribution. Basically, given a set of descriptors extracted from a single image, FV encodes how these descriptors deviate from the trained GMM parameters. Fisher Vectors represent the gradients of the log-likelihood function with respect to the GMM parameters. Formally, given an image with descriptors $X = \{x_t\}_{t=1}^T$ where λ is a vector of GMM parameters and u_λ is a probability density function modeling the distribution of the data.

This gradient is formally known as the score function[9]. It measures the sensitivity of the log-likelihood to changes in the model parameters, essentially indicating how to adjust the GMM to better fit the observed features.

$$G_\lambda^X = L_\lambda \log u_\lambda(X), \quad (4.2)$$

When constructing a FV for an image, a set of local features is assumed to be independent which means that the final descriptor can be constructed as a sum of Fisher Vectors for each local feature:[9]

$$G_\lambda^X = \sum_{t=1}^T L_\lambda \nabla_\lambda \log u_\lambda(x_t) \quad (4.3)$$

The matrix L_λ is derived from the Fisher Information Matrix and is used to "whiten" or to normalize the gradients[21], making them more discriminative and suitable for use in classification later on.

Following the standard practice, this vector is computed separately for means and variances, resulting in a final vector of dimension $2KD$, where K is the number of Gaussian components and D the dimensionality of the PCA-reduced descriptors. L2 and

power normalization have also been applied since they have been shown[22] to generally improve the performance of the method.

FV have also been chosen because they have been shown to perform better than bag-of-words approaches.[20]

4.3.3 Parameter Choices and Rationale

Several key parameters influenced the effectiveness of Fisher Vectors for the re-identification task during the development:

- **Number of PCA components (N_{PCA}):** A balance between dimensionality reduction and retention of discriminative information.
- **Number of GMM components (N_{GMM}):** Controls the granularity of the visual codebook. More components typically yield richer representations but increase computational complexity.

These parameters were empirically determined based on experimenting and computational constraints. Specifically, the PCA dimension was selected to retain approximately 95% of descriptor variance. The number of GMM components was chosen to balance accuracy and computational efficiency. The best results were achieved, with 96 PCA components, 256 GMM components.

4.4 Geometric Verification

Geometric verification is used to make sure that matched keypoints between images truly correspond to the same physical locations on an animal reducing false positives that arise from visual similarities alone. This step is crucial in refining the initial matches obtained through descriptor-based feature matching by enforcing spatial consistency.[9]

4.4.1 Theoretical Foundation

Local feature descriptors such as DISK[3] or KeyNet combined with AffNet and HardNet [23, 24, 25], offer robust initial matches based on visual similarity. However, matches that are generated purely from descriptor similarity may include mistakes, especially

in visually cluttered or repetitive environments common in wildlife images. Geometric verification, as suggested in [9], addresses this by validating matches through spatial coherence checks.

4.4.2 Establishing Initial Correspondences

Initial correspondences are established through descriptor matching using LightGlue[4], an attention-based method. Initially methods like Lowe’s ratio test were considered but matching with LightGlue proved significantly faster and more accurate, effectively eliminating the need for traditional filtering methods.

4.4.3 Transformation Estimation

The core of GV involves estimating a homography transformation between matched key-points using RANSAC (Random Sample Consensus). RANSAC iteratively samples minimal subsets of correspondences, computes candidate transformations and evaluates their agreement with the remaining matches. Matches that fall within a defined spatial threshold are considered inliers.

4.4.4 Scoring and Verification

Matches are assessed primarily by their inlier count, the number of matches consistent with the estimated homography, and their spatial distribution:

- **Inlier count:** Number of correspondences that agree with the estimated transformation; higher counts suggest more reliable matches.
- **Inlier ration:** Normalizes the inlier count by the total number of tentative matches,

$$r = \frac{|I|}{|C|},$$

where $|I|$ denotes the inlier count and $|C|$ the total correspondences.

- **Combined scoring:** Integrates geometric consistency with descriptor similarity (fisher vector distances or global distances) to leverage both spatial and appearance information.

Penalties for low inlier counts or poor geometric coherence discourage accepting borderline matches, improving the overall system reliability and accuracy.

5 Classification

This chapter describes the classification (individual identification) pipeline used in this thesis. The goal is to assign an identity label to a query image by comparing it to a labeled gallery of training images. The full configuration combines three complementary signals: global deep embeddings, aggregated local descriptors (Fisher Vectors 4.3.2) and geometric verification 4.4. It's important to note that the system is modular, meaning that each component can be enabled or disabled, and all of the configurations were individually evaluated 7.

5.1 Problem Formulation

Given a set of labeled training images (the gallery) and an unlabeled query image, the task is to predict the identity of the query by finding the most similar individual in the gallery. Even though this is not realistic in real-world scenarios, in this thesis classification is evaluated in a closed-set protocol, meaning that test identities are restricted to those observed in training. Performance is reported with Top-1 accuracy. Top-N accuracy and F1-Score.

5.2 Overview: a 3-tier funnel

The classification system follows a **retrieve** → **rerank** → **verify** funnel inspired by WildFusion[17] but with some modifications. The key motivation is efficiency since the most expensive stage (geometric verification) is applied only to a shortlist of high-scoring candidates, while fast global/fisher similarities provide that broad shortlist.

Depending on the enabled signals, the same structure can represent several baselines: (i) global-only retrieval, (ii) FV-only retrieval (inspired by ALFRED Re-ID pipeline[9])

and (iii) the full global + FV + GV fusion.

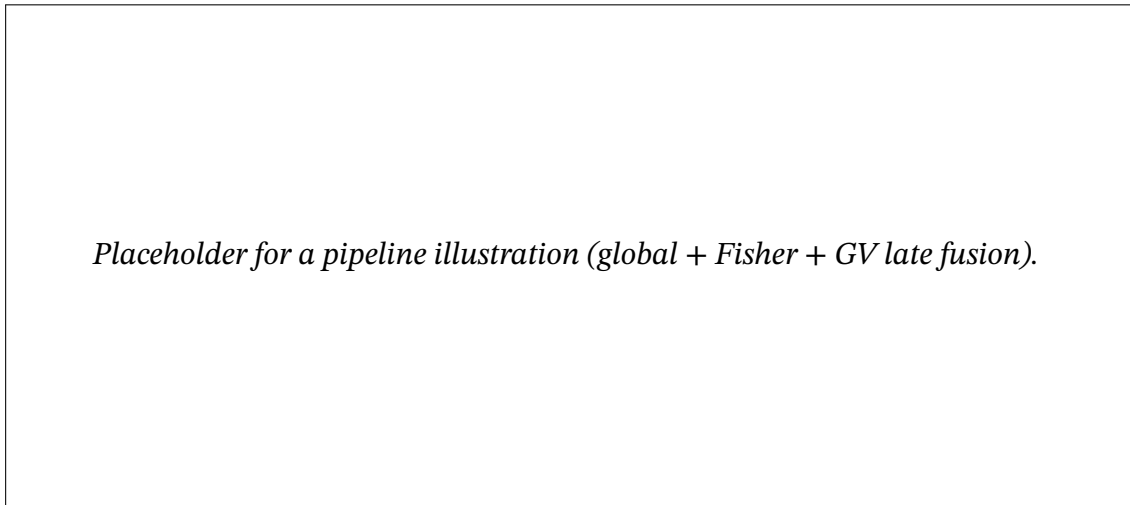


Figure 5.1: High-level overview of the modular classification pipeline. The figure will illustrate the three tiers: (1) union retrieval from global embeddings and Fisher vectors, (2) calibrated late fusion reranking, and (3) geometric verification based reranking.

The three tiers are:

1. **Tier 1 (candidate retrieval):** retrieve top candidates using global embedding similarity and FV similarity, then take the union of those two sets
2. **Tier 2 (calibrated late fusion):** calibrate each similarity score to a probability of being the same individual and combine them into a single ranking score
3. **Tier 3 (GV reranking):** compute a geometric consistency signal (inlier count) and use it to push up the ranking of the candidates that are both visually similar and geometrically plausible.

5.3 Tier 1: Candidate Retrieval

Tier 1 produces a high-recall shortlist cheaply. As mentioned before, two complementary representations are used:

- **Global embeddings**4.2.2: a single vector per image, compared against others using cosine similarity and extracted using a pre-trained MegaDescriptor-L-384 model

- **Fisher vectors**^{4.3.2}: aggregated local descriptors that retain fine-grained texture information, motivated by the ALFRED Re-ID pipeline[9]

For a query image, the system computes cosine similarities to all gallery images separately for the modalities that are turned on (global, FV) and selects the top B candidates from each list (in this thesis that parameter B was set to 500). The union of these sets forms the Tier 1 candidate pool. The idea behind this union strategy was to reduce the risk of missing the true match when one modality fails. If either FV or global embeddings are turned off, then naturally there is no union and the candidate pool is entirely produced by the modality that is turned on.

5.4 Tier 2: Score Calibration and Fusion

Raw similarity values from different modalities are not directly comparable. They have different scales, different score distributions etc. Following the rationale in WildFusion[17] this thesis applies **per-dataset similarity calibration** to map each raw similarity into an estimated probability of being the same individual:

$$p_s(x, y) \approx P(\text{same} \mid s(x, y)), \quad (5.1)$$

where $s(x, y)$ is a raw similarity score for a pair of images (x, y) in a given modality.

5.4.1 Calibration Set Construction

Calibration requires labeled pairs. For each dataset, a subset of training images is sampled as calibration queries. Positive pairs are formed by matching each query with other images of the same identity. Negative pairs should be representative of hard confusions so when global embeddings are available, negatives are sampled from the top- B global shortlist of the query, which tends to produce visually similar non-matches.[17]

5.4.2 Calibrators and Fusion Rules

Each active Tier 2 signal (global and/or FV) is calibrated independently. In practice, this is implemented as a monotonic calibration model (isotonic regression) which has been

shown to work well with small datasets[17]. The calibrated probabilities are then fused by a simple mean:

$$p_{\text{tier2}}(x, y) = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} p_s(x, y), \quad (5.2)$$

where $\mathcal{S} \subseteq \{\text{global, fisher}\}$ is the set of enabled Tier 2 signals. This produces a single probabilistic similarity estimate used to rank the Tier 1 union candidates.

5.5 Tier 3: Geometric Verification Reranking

Even strong appearance similarity can produce false positives, especially for visually similar individuals. GV addresses this by enforcing spatial consistency of matched keypoints, as described in [9]. In this thesis, tentative matches are obtained using LightGlue and a geometric model is estimated with RANSAC/MAGSAC. The number of inliers () serves as a measure of geometric consistency.

Because GV is computationally expensive, it is applied only to a small number of top-ranked Tier 2 candidates. In the thesis this was set to 200. If $n(x, y)$ is the inlier count between query x and candidate y , the Tier 2 probabilities are converted to distance $d_L = 1 - p_{\text{tier2}}$ and then adjusted with a power rule[9]:

$$d_C(x, y) = d_L(x, y)^{n(x, y)}. \quad (5.3)$$

Intuition behind is that, since the distance d_L is a fraction between 0 and 1, raising it to a power makes it smaller. This creates an effect by which when there is a strong geometry (n is large), d_L is raised to a higher power, causing the distance d_C to fall towards zero. This boosts the geometrically verified matches towards the top of the list. On the other hand, if no keypoints match ($n = 0$), the geometric term provides no boost ($d_L^0 = 1$) and in these cases the system falls back to the original Tier 2 ranking.

5.6 End-to-end Inference

Putting this all together, the classification goes like this:

1. Extract global embeddings and/or local features and compute FVs.
2. Train the calibrators on a small labeled subset of training data.
3. For each query image:
 - (a) Retrieve a Tier 1 union shortlist from global and fisher similarity.
 - (b) Compute calibrated Tier 2 probabilities and rank by p_{tier2} .
 - (c) Apply GV to the top candidates and rerank by the GV power rule.
4. Output the predicted identity (Top-1) and the Top- N ranked identities for evaluation.

This design was created to provide a practical trade-off between accuracy and compute: global and fisher signals offer fast retrieval (after the initial caching, more on this will be said in later chapters) and strong discrimination, while GV acts as a fine filter for difficult cases where similarity alone is not good enough.

6 Population Size Estimation

This chapter describes the population size estimation component of the thesis. The goal is to estimate the number of unique individuals in a large image collection using the re-identification methods introduced in previous chapters while avoiding exhaustive pairwise comparison between all images.

6.1 Introduction

While individual animal re-identification provides the foundation for wildlife monitoring, the ultimate goal in many conservation and ecological applications is estimating population sizes. Traditional clustering approaches for population estimation often produce inaccurate results when applied to re-identification systems, particularly when the underlying pairwise similarities are noisy or unreliable [5].

Perez et al.[5] proposed a statistically focused alternative: *Human-in-the-Loop Nested Importance Sampling* (HITL-NIS). Instead of attempting to fully cluster the dataset, the method estimates the cluster count directly by adaptively sampling a small number of image pair for vetting by an "oracle" (a human annotator that decides whether a pair of images is of a same individual). For the purposes of their research and for the purposes of this thesis as well, human annotation was simulated by just taking the GT values from the publicly available datasets on which this was tested. Since the original paper didn't address the very real concern of human error during annotation, in this thesis I introduced a simulated variable of *human labeling error* to test how it affects the estimator. The results of this analysis are presented in the next chapter, in this chapter I'll mostly focus on the theoretical background and implementation.

6.2 Theoretical Foundation

6.2.1 Population Size as Connected Components

The population estimation problem can be formulated as counting connected components in a graph where vertices represent images and edges exist between images of the same individual. Formally, given a graph $G = (V, E)$ where vertices correspond to images and edges represent same-individual relationships, the number of clusters K equals the number of connected components:

$$K = \text{CC}(G) = \sum_{u=1}^n \frac{1}{1 + d(u)} \quad (6.1)$$

where $d(u)$ is the degree of vertex u (i.e., the number of edges connected to u) [5].

This elegant formulation arises from the observation that the graph G consists of a collection of cliques, where each clique represents images of the same individual. The clique containing vertex u has size $1 + d(u)$, and the total contribution of all vertices in this clique is $(1 + d(u)) \times \frac{1}{1 + d(u)} = 1$. Thus, each clique contributes exactly 1 to the sum, yielding the total number of connected components.

This formulation provides the foundation for statistical estimation: instead of computing all pairwise similarities explicitly, we can sample vertices and estimate their degrees, then use importance sampling to obtain an unbiased estimate of the population size.

6.2.2 Two-Stage Estimation: Nested Monte Carlo and Nested Importance Sampling

The most direct Monte Carlo approach samples vertices uniformly:

$$\text{CC}(G) = n \mathbb{E}_{u \sim \text{Unif}(\{1, \dots, n\})} \left[\frac{1}{1 + d(u)} \right]. \quad (6.2)$$

For a sampled vertex u , its degree can be estimated by sampling candidate neighbors v the annotator whether an edge exists (are they the same individual in this case). Nested

Recommended Figure. HITL-NIS overview: outer vertex sampling $\rightarrow$ inner neighbor proposal $\rightarrow$ oracle query $\rightarrow$ importance-weighted degree estimate $\hat{d}(u)$ $\rightarrow$ component-count contribution $\rightarrow$ average + confidence interval.

Figure 6.1: Schematic of Human-in-the-Loop Nested Importance Sampling used for population-size estimation.

Monte Carlo (Nested-MC) does this with uniform neighbor sampling. Nested importance sampling (NIS) replaces uniform sampling with proposal distributions designed to reduce variance. The paper suggests an outer proposal Q that oversamples low-degree vertices and an inner proposal q that oversamples likely neighbors[5].

In it's general form, NIS estimator can be written like:

$$\widehat{CC}_{NIS}(G) = \frac{1}{N} \sum_{i=1}^N \left(\frac{1}{Q(u_i)} \cdot \frac{1}{1 + \hat{d}(u_i)} \right), \quad u_i \sim Q(u). \quad (6.3)$$

The degree estimator for a sampled vertex u uses importance weighting over neighbors:

$$\hat{d}(u) = \frac{1}{M} \sum_{j=1}^M \frac{s(u, v_j)}{q_u(v_j)}, \quad v_j \sim q_u(v), \quad (6.4)$$

where $s(u, v) \in \{0, 1\}$ is the *true* edge indicator. In HITL settings, $s(u, v)$ is revealed by a human annotator and in this thesis it is simulated from ground-truth values during experiments.

6.3 HITL-NIS Implementation

This section describes how population counting was implemented in this thesis and how it differs from the description in the original paper.

6.3.1 Graph and Annotator Abstraction

The counting pipeline represents the dataset by a list of image identifier $\{id_i\}_{i=1}^n$. The graph G is never constructed explicitly, instead the algorithm queries an "annotator"

function $\mathcal{O}(id_u, id_v) \in \{0, 1\}$ that returns whether two images belong to the same individual. As mentioned before, in the experiments for this thesis, $\mathcal{O}$ is simulated using ground-truth identity labels.

6.3.2 Cheap Proposal Signals: Global Embeddings and Fisher Vectors

Human annotation is "expensive" since it requires a human in real development, so the method relies on cheap embedding signals to prioritize which pairs to query. As with classification, two base signals are supported.

- **Global embedding similarity:** a single global feature vector is extracted per image (MD) and after L2 normalization cosine similarity is computed
- **FV similarity:** local descriptors are aggregated into FVs as described in Chapter 4.3 and again after L2 normalization, cosine similarity is computed

These signals are used *only* for proposal distributions, the estimator ultimately depends on annotator verified edge indicators.

6.3.3 Calibrating Scores to Probabilities for Sampling

Raw cosine similarities are not calibrated and aren't comparable across signals. To obtain a proposal distribution that is closer to the ideal $q_u(v) \propto s(u, v)$, the implementation support per signal calibration $s_{\text{raw}}(u, v) \mapsto \hat{p}(\text{same} \mid s_{\text{raw}})$ using monotone calibrators (isotonic regression, optionally with PCHIP interpolation for tie handling). Calibration pairs are constructed from GT identities (this, ofcourse, means that it increases the human effort needed to label the calibration pairs, but only by about 1000-2000 pairs).

To better match the distribution of pairs encountered during counting, negative calibration pairs are samples as hard negatives from a shortlist of high-similarity candidates, rather than uniformly at random. This is important because the decision boundary between "same" and "different" is most important among the near neighbors where the model is uncertain.

At runtime, calibrated probabilities from the available signals are fused by averaging:

$$\hat{p}_{\text{base}}(u, v) = \frac{1}{S} \sum_{m=1}^S \hat{p}_m(u, v), \quad (6.5)$$

where $S \in \{1, 2\}$ depending on whether global embeddings and/or Fisher vectors are enabled.

6.3.4 Outer Sampling Choice

The original paper proposes an importance-weighted outer distribution $Q(u) \propto 1/(1 + \hat{d}(u))$, where $\hat{d}(u)$ is estimated from approximate similarities [5]. This can reduce estimator variance by oversampling low-degree vertices that contribute more to $\text{CC}(G)$.

In this thesis, the outer sampling is set to **uniform** $Q(u) = 1/n$. The motivation is that constructing a reliable approximation to $Q(u)$ would require estimating $\sum_{v \neq u} \hat{s}(u, v)$ for *all* u , which can approach $O(n^2)$ computation/memory. By keeping Q uniform, the method retains the same unbiased estimator structure while focusing variance reduction on the inner neighbor proposal q which delivers the main savings in human effort.

6.3.5 Complexity and Diagnostics

The total human query budget is $N \times M$ (plus optional confirmation votes; see Section 6.4). For each run, the implementation reports: the estimate $\hat{K}$ and a 95% confidence interval.

6.4 Human Labeling Error

HITL-NIS assumes that a human annotator correctly labels each of the presented pairs. It's self evident that in practice this is a very wrong assumption. Human judgments will always be imperfect, especially for visually similar individuals and if the annotator is someone that's generally not very familiar with the animal species in question. This is not just a small issue, this directly violates the clique-graph assumption behind equation 6.1 because it introduces wrong edges. This is very harmful. Pairwise comparisons in a large dataset are dominated by non-matches, this means that even a small FP rate (incorrect labeling) can introduce made up edges that inflate $\hat{d}(u)$. Since the estimator

uses $1/(1 + \hat{d}(u))$, such errors can drastically decrease the estimated count (bias the population estimate downward by effectively merging identities). This bias is not captured by confidence intervals since that accounts only for Monte Carlo sampling variance. This was not at all addressed in the original paper and I'll now discuss how I attempted to address it.

6.4.1 Explicit Noise Modeling

To quantify the robustness of the method, I simulated a human error parameter by exposing the system to a parameter that introduces a random error in ranges from 2% to 30%. As expected, without any method to address this, the results of counting become completely unreliable at a 2% error rate already. These results are reported in the next chapter.

6.4.2 Confirmation by Repeated Voting

False positives are more damaging than false negatives for cluster counting. To reduce false positives while limiting extra human annotation cost, a following confirmation strategy was implemented:

If the first human vote is “different”, accept it immediately. If the first vote is “same”, re-query the *same pair* and accept “same” only if it is returned K times consecutively.

This method adds cost only for candidate matches (which are relatively rare), while reducing the probability that a single mistaken “same” response merges identities wrongly. The trade-off between robustness and additional human annotator calls is evaluated in the next chapter.

6.5 Parameter Configuration

The implementation exposes parameters that control estimator variance, query budget, and robustness:

- N : number of sampled outer-loop vertices

- M : number of sampled neighbors per vertex
- **Simulated label error**: human mistake probability p .
- **Positive confirmation**: number of consecutive “same” votes required to accept a match.

Increasing N and M reduces estimator variance but increases human annotator calls. Increasing the confirmation parameter improves robustness to false positives but also increases human annotator calls.

7 Experiments and Results

- Rezultati klasifikacije (top-1, top-n acc), rezultati prebrojavanja individualnih životinja, rezultati prebrojavanja na mom datasetu
- Vizualizacije, tablice
- Usporedba utjecaja hiperparametara testiranih

8 Conclusion and Future Work

- Summary of contributions and findings.
- Potential extensions and open questions.

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Abstract

Deep learning-based animal re-identification for non-invasive wildlife monitoring and conservation

Matej Marić

Reliable methods for monitoring wildlife populations are essential for effective conservation and ecological management. This thesis addresses the challenge of developing an automated visual system capable of identifying individual animals and estimating their population size across various species, particularly when limited labeled data is available. The primary goal was to establish a practical approach that can generalize effectively without the need for extensive manual annotation or species-specific model retraining. The developed methods were initially validated on publicly available benchmark datasets and subsequently tested using original data collected through trail cameras placed on Medvednica mountain in Zagreb, Croatia. The research specifically aimed at evaluating the system's practical applicability and accuracy in real-world conditions, focusing on local populations of wild boars (*Sus scrofa*) and roe deer (*Capreolus capreolus*).

Keywords: Animal Re-identification; Wildlife Monitoring; Population Estimation; Computer Vision; Conservation

Sažetak

Duboko učenje za ponovnu identifikaciju životinja u neinvazivnom praćenju divljih vrsta i očuvanju prirode

Matej Marić

Pouz dane metode praćenja populacija divljih životinja ključne su za učinkovito očuvanje i ekološko upravljanje. Ovaj rad bavi se izazovom razvoja automatiziranog vizualnog sustava sposobnog za identificiranje pojedinačnih životinja i procjenu veličine njihove populacije među različitim vrstama, posebno kada je dostupa ograničen broj označenih podataka. Primarni cilj bio je uspostaviti praktičan pristup koji se može učinkovito generalizirati bez potrebe za opsežnim ručnim označavanjem ili ponovnim treniranjem modela specifičnih za vrstu. Razvijene metode u početku su validirane na javno dostupnim skupovima podataka, a potom testirane korištenjem originalnih podataka prikupljenih putem kamera na stazama postavljenih na Medvednici u Zagrebu, Hrvatska. Istraživanje je posebno bilo usmjereno na procjenu praktične primjenjivosti i točnosti sustava u stvarnim uvjetima, s naglaskom na lokalne populacije divljih svinja (*Sus scrofa*) i srna (*Capreolus capreolus*).

Ključne riječi: Re-identifikacija životinja; Nadzor Divljači; Procjena Populacije; Računalni Vid; Očuvanje Prirode